

Stage 3 (all parts together): $\mathcal{L} = \mathcal{L}_{\text{fwd}} + \mathcal{L}_{\text{inv}} + \mathcal{L}_{\text{dir}, a} + \mathcal{L}_{\text{dir}, y} + \mathcal{L}_{\text{VAE}}$

Stage 1 (invertible blocks only): $\mathcal{L} = \mathcal{L}_{\text{fwd}} + \mathcal{L}_{\text{inv}} + \mathcal{L}_{\text{dir}, a} + \mathcal{L}_{\text{dir}, y}$

Stage 2 (VAE only): $\mathcal{L} = \mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \beta \mathcal{L}_{\text{KL}}, \quad \beta = 0.05$

$$\mathcal{L}_{\text{fwd}} = \text{mean} (\hat{y} - y)^2, \quad \mathcal{L}_{\text{inv}} = \text{mean} (\hat{a} - a)^2$$

$$\mathcal{L}_{\text{dir}, y} = \text{mean} (\tilde{y} - y)^2, \quad \mathcal{L}_{\text{dir}, a} = \text{mean} (\tilde{a} - a)^2, \quad \mathcal{L}_{\text{recon}} = \text{mean} (a_{\text{rec}} - a)^2$$

a : design; y : ERP; $\mathcal{L}_{\text{KL}}$: distance of the VAE latent distribution from $\mathcal{N}(0, 1)$

$\hat{y}$, $\hat{a}$: ERP and design predicted through the invertible blocks (forward and inverse)

$\tilde{y}$, $\tilde{a}$: ERP and design read out directly from the lifted ERP and design (blocks skipped)

a_{rec} : design reconstructed by the VAE